

Filter Summary Report: CG,TIA,Automated,simple,Z1,Z2

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Contents

1 Examined $H(z)$ for CG TIA Automated simple **Z1 Z2:** $Z_1Z_2g_m + Z_1$

$$H(z) = Z_1Z_2g_m + Z_1$$

2 AP

3 BP

3.1 **BP-1** $Z(s) = \left(\frac{L_1R_1s}{C_1L_1R_1s^2+L_1s+R_1}, \ R_2, \ \infty, \ \infty, \ \infty, \ \infty \right)$

$$H(s) = \frac{s \left(L_1R_1R_2g_m + L_1R_1 \right)}{C_1L_1R_1s^2 + L_1s + R_1}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_1}{\sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{1}{C_1R_1}$
K-LP: 0
K-HP: 0
K-BP: $R_1R_2g_m + R_1$
Qz: None
Wz: None

4 BS

4.1 **BS-1** $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \ R_2, \ \infty, \ \infty, \ \infty, \ \infty \right)$

$$H(s) = \frac{R_1R_2g_m + R_1 + s^2 \left(C_1L_1R_1R_2g_m + C_1L_1R_1 \right)}{C_1L_1s^2 + C_1R_1s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_1}}{\sqrt{C_1}R_1}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{R_1}{L_1}$
K-LP: $R_1R_2g_m + R_1$
K-HP: $R_1R_2g_m + R_1$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

5 GE

5.1 **GE-1** $Z(s) = \left(R_1, \ \frac{R_2(C_2L_2s^2+1)}{C_2L_2s^2+C_2R_2s+1}, \ \infty, \ \infty, \ \infty, \ \infty \right)$

$$H(s) = \frac{C_2R_1R_2s + R_1R_2g_m + R_1 + s^2 \left(C_2L_2R_1R_2g_m + C_2L_2R_1 \right)}{C_2L_2s^2 + C_2R_2s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_2}}{\sqrt{C_2}R_2}$
wo: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$

bandwidth: $\frac{R_2}{L_2}$
K-LP: $R_1 R_2 g_m + R_1$
K-HP: $R_1 R_2 g_m + R_1$
K-BP: R_1
Qz: $\frac{\sqrt{L_2 R_2 g_m} + \sqrt{L_2}}{\sqrt{C_2 R_2}}$
Wz: $\frac{1}{\sqrt{C_2 \sqrt{L_2}}}$

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_2 s + R_1 R_2 g_m + R_1}{C_1 C_2 R_1 R_2 s^2 + s (C_1 R_1 + C_2 R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2}}{C_1 R_1 + C_2 R_2}$
wo: $\frac{1}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2}}$
bandwidth: $\frac{C_1 R_1 + C_2 R_2}{C_1 C_2 R_1 R_2}$
K-LP: $R_1 R_2 g_m + R_1$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_2}{C_1 R_1 + C_2 R_2}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_2 L_1 R_1 s + L_1 R_1 g_m}{C_1 C_2 L_1 R_1 s^2 + C_2 L_1 s + C_2 R_1}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
bandwidth: $\frac{1}{C_1 R_1}$
K-LP: $\frac{L_1 g_m}{C_2}$
K-HP: 0
K-BP: R_1
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{L_1 R_1 g_m + s (C_2 L_1 R_1 R_2 g_m + C_2 L_1 R_1)}{C_1 C_2 L_1 R_1 s^2 + C_2 L_1 s + C_2 R_1}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
bandwidth: $\frac{1}{C_1 R_1}$

K-LP: $\frac{L_1 g_m}{C_2}$
K-HP: 0
K-BP: $R_1 R_2 g_m + R_1$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, R_2, \infty, \infty, \infty, \infty)$

$$H(s) = R_1 R_2 g_m + R_1$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty\right)$

$$H(s) = \frac{C_2 R_1 s + R_1 g_m}{C_2 s}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty\right)$

$$H(s) = \frac{C_2 R_1 R_2 s + R_1 R_2 g_m + R_1}{C_2 R_2 s + 1}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty\right)$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_2 s}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty\right)$

$$H(s) = \frac{C_2 L_2 R_1 g_m s^2 + C_2 R_1 s + R_1 g_m}{C_2 s}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, L_2 s + R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty\right)$

$$H(s) = \frac{C_2 L_2 R_1 g_m s^2 + R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_2 s}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty\right)$

$$H(s) = \frac{L_2 R_1 g_m s + R_1 R_2 g_m + R_1 + s^2 (C_2 L_2 R_1 R_2 g_m + C_2 L_2 R_1)}{C_2 L_2 s^2 + 1}$$

9.8 X-INVALID-ORDER-8 $Z(s) = (L_1 s, R_2, \infty, \infty, \infty, \infty)$

$$H(s) = s (L_1 R_2 g_m + L_1)$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(L_1 s, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 s + L_1 g_m}{C_2}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(L_1 s, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 R_2 s^2 + s (L_1 R_2 g_m + L_1)}{C_2 R_2 s + 1}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(L_1 s, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_1 g_m + s (C_2 L_1 R_2 g_m + C_2 L_1)}{C_2}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(L_1 s, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 L_2 g_m s^2 + C_2 L_1 s + L_1 g_m}{C_2}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(L_1 s, L_2 s + R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 L_2 g_m s^2 + L_1 g_m + s (C_2 L_1 R_2 g_m + C_2 L_1)}{C_2}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(L_1 s, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_1 L_2 g_m s^2 + s^3 (C_2 L_1 L_2 R_2 g_m + C_2 L_1 L_2) + s (L_1 R_2 g_m + L_1)}{C_2 L_2 s^2 + 1}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(L_1 s, \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 R_2 s^2 + s^3 (C_2 L_1 L_2 R_2 g_m + C_2 L_1 L_2) + s (L_1 R_2 g_m + L_1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, R_2, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + 1}{C_1 s}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 s + g_m}{C_1 C_2 s^2}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 R_2 s + R_2 g_m + 1}{C_1 C_2 R_2 s^2 + C_1 s}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{g_m + s (C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad L_2 s + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 L_2 g_m s^2 + C_2 s + g_m}{C_1 C_2 s^2}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 L_2 g_m s^2 + g_m + s (C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_2 g_m s + R_2 g_m + s^2 (C_2 L_2 R_2 g_m + C_2 L_2) + 1}{C_1 C_2 L_2 s^3 + C_1 s}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 R_2 s + R_2 g_m + s^2 (C_2 L_2 R_2 g_m + C_2 L_2) + 1}{C_1 C_2 L_2 s^3 + C_1 C_2 R_2 s^2 + C_1 s}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_2 g_m + R_1}{C_1 R_1 s + 1}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 R_1 s + R_1 g_m}{C_1 C_2 R_1 s^2 + C_2 s}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_1 C_2 R_1 s^2 + C_2 s}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad L_2 s + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 L_2 R_1 g_m s^2 + C_2 R_1 s + R_1 g_m}{C_1 C_2 R_1 s^2 + C_2 s}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 L_2 R_1 g_m s^2 + R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_1 C_2 R_1 s^2 + C_2 s}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_2 R_1 g_m s + R_1 R_2 g_m + R_1 + s^2 (C_2 L_2 R_1 R_2 g_m + C_2 L_2 R_1)}{C_1 C_2 L_2 R_1 s^3 + C_1 R_1 s + C_2 L_2 s^2 + 1}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 R_1 R_2 s + R_1 R_2 g_m + R_1 + s^2 (C_2 L_2 R_1 R_2 g_m + C_2 L_2 R_1)}{C_1 C_2 L_2 R_1 s^3 + s^2 (C_1 C_2 R_1 R_2 + C_2 L_2) + s (C_1 R_1 + C_2 R_2) + 1}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s (C_1 R_1 R_2 g_m + C_1 R_1) + 1}{C_1 s}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 R_1 s^2 + g_m + s (C_1 R_1 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 R_1 R_2 s^2 + R_2 g_m + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2) + 1}{C_1 C_2 R_2 s^2 + C_1 s}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_2 R_1 g_m s^3 + g_m + s^2 (C_1 C_2 R_1 + C_2 L_2 g_m) + s (C_1 R_1 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, L_2 s + R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_2 R_1 g_m s^3 + g_m + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_2 L_2 g_m) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^3 (C_1 C_2 L_2 R_1 R_2 g_m + C_1 C_2 L_2 R_1) + s^2 (C_1 L_2 R_1 g_m + C_2 L_2 R_2 g_m + C_2 L_2) + s (C_1 R_1 R_2 g_m + C_1 R_1 + L_2 g_m) + 1}{C_1 C_2 L_2 s^3 + C_1 s}$$

$$\begin{aligned}
\text{9.38} \quad \text{X-INVALID-ORDER-38} \quad Z(s) &= \left(R_1 + \frac{1}{C_1 s}, \frac{R_2(C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{R_2 g_m + s^3 (C_1 C_2 L_2 R_1 R_2 g_m + C_1 C_2 L_2 R_1) + s^2 (C_1 C_2 R_1 R_2 + C_2 L_2 R_2 g_m + C_2 L_2) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2) + 1}{C_1 C_2 L_2 s^3 + C_1 C_2 R_2 s^2 + C_1 s}
\end{aligned}$$

$$\begin{aligned}
\text{9.39} \quad \text{X-INVALID-ORDER-39} \quad Z(s) &= \left(L_1 s + \frac{1}{C_1 s}, R_2, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{R_2 g_m + s^2 (C_1 L_1 R_2 g_m + C_1 L_1) + 1}{C_1 s}
\end{aligned}$$

$$\begin{aligned}
\text{9.40} \quad \text{X-INVALID-ORDER-40} \quad Z(s) &= \left(L_1 s + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{C_1 C_2 L_1 s^3 + C_1 L_1 g_m s^2 + C_2 s + g_m}{C_1 C_2 s^2}
\end{aligned}$$

$$\begin{aligned}
\text{9.41} \quad \text{X-INVALID-ORDER-41} \quad Z(s) &= \left(L_1 s + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{C_1 C_2 L_1 R_2 s^3 + C_2 R_2 s + R_2 g_m + s^2 (C_1 L_1 R_2 g_m + C_1 L_1) + 1}{C_1 C_2 R_2 s^2 + C_1 s}
\end{aligned}$$

$$\begin{aligned}
\text{9.42} \quad \text{X-INVALID-ORDER-42} \quad Z(s) &= \left(L_1 s + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{C_1 L_1 g_m s^2 + g_m + s^3 (C_1 C_2 L_1 R_2 g_m + C_1 C_2 L_1) + s (C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}
\end{aligned}$$

$$\begin{aligned}
\text{9.43} \quad \text{X-INVALID-ORDER-43} \quad Z(s) &= \left(L_1 s + \frac{1}{C_1 s}, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{C_1 C_2 L_1 L_2 g_m s^4 + C_1 C_2 L_1 s^3 + C_2 s + g_m + s^2 (C_1 L_1 g_m + C_2 L_2 g_m)}{C_1 C_2 s^2}
\end{aligned}$$

$$\begin{aligned}
\text{9.44} \quad \text{X-INVALID-ORDER-44} \quad Z(s) &= \left(L_1 s + \frac{1}{C_1 s}, L_2 s + R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{C_1 C_2 L_1 L_2 g_m s^4 + g_m + s^3 (C_1 C_2 L_1 R_2 g_m + C_1 C_2 L_1) + s^2 (C_1 L_1 g_m + C_2 L_2 g_m) + s (C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}
\end{aligned}$$

$$\begin{aligned}
\text{9.45} \quad \text{X-INVALID-ORDER-45} \quad Z(s) &= \left(L_1 s + \frac{1}{C_1 s}, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{C_1 L_1 L_2 g_m s^3 + L_2 g_m s + R_2 g_m + s^4 (C_1 C_2 L_1 L_2 R_2 g_m + C_1 C_2 L_1 L_2) + s^2 (C_1 L_1 R_2 g_m + C_1 L_1 + C_2 L_2 R_2 g_m + C_2 L_2) + 1}{C_1 C_2 L_2 s^3 + C_1 s}
\end{aligned}$$

$$\begin{aligned}
\text{9.46} \quad \text{X-INVALID-ORDER-46} \quad Z(s) &= \left(L_1 s + \frac{1}{C_1 s}, \frac{R_2(C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right) \\
H(s) &= \frac{C_1 C_2 L_1 R_2 s^3 + C_2 R_2 s + R_2 g_m + s^4 (C_1 C_2 L_1 L_2 R_2 g_m + C_1 C_2 L_1 L_2) + s^2 (C_1 L_1 R_2 g_m + C_1 L_1 + C_2 L_2 R_2 g_m + C_2 L_2) + 1}{C_1 C_2 L_2 s^3 + C_1 C_2 R_2 s^2 + C_1 s}
\end{aligned}$$

$$\mathbf{9.47 \quad X-INVALID-ORDER-47} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, R_2, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{s(L_1 R_2 g_m + L_1)}{C_1 L_1 s^2 + 1}$$

$$\mathbf{9.48 \quad X-INVALID-ORDER-48} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 s + L_1 g_m}{C_1 C_2 L_1 s^2 + C_2}$$

$$\mathbf{9.49 \quad X-INVALID-ORDER-49} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 R_2 s^2 + s(L_1 R_2 g_m + L_1)}{C_1 C_2 L_1 R_2 s^3 + C_1 L_1 s^2 + C_2 R_2 s + 1}$$

$$\mathbf{9.50 \quad X-INVALID-ORDER-50} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_1 g_m + s(C_2 L_1 R_2 g_m + C_2 L_1)}{C_1 C_2 L_1 s^2 + C_2}$$

$$\mathbf{9.51 \quad X-INVALID-ORDER-51} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 L_2 g_m s^2 + C_2 L_1 s + L_1 g_m}{C_1 C_2 L_1 s^2 + C_2}$$

$$\mathbf{9.52 \quad X-INVALID-ORDER-52} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, L_2 s + R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 L_2 g_m s^2 + L_1 g_m + s(C_2 L_1 R_2 g_m + C_2 L_1)}{C_1 C_2 L_1 s^2 + C_2}$$

$$\mathbf{9.53 \quad X-INVALID-ORDER-53} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_1 L_2 g_m s^2 + s^3(C_2 L_1 L_2 R_2 g_m + C_2 L_1 L_2) + s(L_1 R_2 g_m + L_1)}{C_1 C_2 L_1 L_2 s^4 + s^2(C_1 L_1 + C_2 L_2) + 1}$$

$$\mathbf{9.54 \quad X-INVALID-ORDER-54} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \frac{R_2(C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 R_2 s^2 + s^3(C_2 L_1 L_2 R_2 g_m + C_2 L_1 L_2) + s(L_1 R_2 g_m + L_1)}{C_1 C_2 L_1 L_2 s^4 + C_1 C_2 L_1 R_2 s^3 + C_2 R_2 s + s^2(C_1 L_1 + C_2 L_2) + 1}$$

$$\mathbf{9.55 \quad X-INVALID-ORDER-55} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, R_2, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^2(C_1 L_1 R_2 g_m + C_1 L_1) + s(C_1 R_1 R_2 g_m + C_1 R_1) + 1}{C_1 s}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 s^3 + g_m + s^2(C_1 C_2 R_1 + C_1 L_1 g_m) + s(C_1 R_1 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 R_2 s^3 + R_2 g_m + s^2 (C_1 C_2 R_1 R_2 + C_1 L_1 R_2 g_m + C_1 L_1) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2) + 1}{C_1 C_2 R_2 s^2 + C_1 s}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_1 C_2 L_1 R_2 g_m + C_1 C_2 L_1) + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 L_2 g_m s^4 + g_m + s^3 (C_1 C_2 L_1 + C_1 C_2 L_2 R_1 g_m) + s^2 (C_1 C_2 R_1 + C_1 L_1 g_m + C_2 L_2 g_m) + s (C_1 R_1 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, L_2 s + R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 L_2 g_m s^4 + g_m + s^3 (C_1 C_2 L_1 R_2 g_m + C_1 C_2 L_1 + C_1 C_2 L_2 R_1 g_m) + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 L_1 g_m + C_2 L_2 g_m) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^4 (C_1 C_2 L_1 L_2 R_2 g_m + C_1 C_2 L_1 L_2) + s^3 (C_1 C_2 L_2 R_1 R_2 g_m + C_1 C_2 L_2 R_1 + C_1 L_1 L_2 g_m) + s^2 (C_1 L_1 R_2 g_m + C_1 L_1 + C_1 L_2 R_1 g_m + C_2 L_2 R_2 g_m + C_2 L_2) + s (C_1 R_1 R_2 g_m + C_1 R_1 + L_2 g_m) + 1}{C_1 C_2 L_2 s^3 + C_1 s}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^4 (C_1 C_2 L_1 L_2 R_2 g_m + C_1 C_2 L_1 L_2) + s^3 (C_1 C_2 L_1 R_2 + C_1 C_2 L_2 R_1 R_2 g_m + C_1 C_2 L_2 R_1) + s^2 (C_1 C_2 R_1 R_2 + C_1 L_1 R_2 g_m + C_1 L_1 + C_2 L_2 R_2 g_m + C_2 L_2) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2) + 1}{C_1 C_2 L_2 s^3 + C_1 C_2 R_2 s^2 + C_1 s}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 R_1 R_2 s^2 + s (L_1 R_1 R_2 g_m + L_1 R_1)}{C_1 C_2 L_1 R_1 R_2 s^3 + R_1 + s^2 (C_1 L_1 R_1 + C_2 L_1 R_2) + s (C_2 R_1 R_2 + L_1)}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_1 L_2 R_1 g_m s^2 + s^3 (C_2 L_1 L_2 R_1 R_2 g_m + C_2 L_1 L_2 R_1) + s (L_1 R_1 R_2 g_m + L_1 R_1)}{C_1 C_2 L_1 L_2 R_1 s^4 + C_2 L_1 L_2 s^3 + L_1 s + R_1 + s^2 (C_1 L_1 R_1 + C_2 L_2 R_1)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 L_1 R_1 R_2 s^2 + s^3 (C_2 L_1 L_2 R_1 R_2 g_m + C_2 L_1 L_2 R_1) + s (L_1 R_1 R_2 g_m + L_1 R_1)}{C_1 C_2 L_1 L_2 R_1 s^4 + R_1 + s^3 (C_1 C_2 L_1 R_1 R_2 + C_2 L_1 L_2) + s^2 (C_1 L_1 R_1 + C_2 L_1 R_2 + C_2 L_2 R_1) + s (C_2 R_1 R_2 + L_1)}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, R_2, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s^2 (C_1 L_1 R_1 R_2 g_m + C_1 L_1 R_1) + s (L_1 R_2 g_m + L_1)}{C_1 L_1 s^2 + 1}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 R_1 s^3 + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_2 L_1) + s (C_2 R_1 + L_1 g_m)}{C_1 C_2 L_1 s^3 + C_2 s}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 R_1 R_2 s^3 + R_1 R_2 g_m + R_1 + s^2 (C_1 L_1 R_1 R_2 g_m + C_1 L_1 R_1 + C_2 L_1 R_2) + s (C_2 R_1 R_2 + L_1 R_2 g_m + L_1)}{C_1 C_2 L_1 R_2 s^3 + C_1 L_1 s^2 + C_2 R_2 s + 1}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^3 (C_1 C_2 L_1 R_1 R_2 g_m + C_1 C_2 L_1 R_1) + s^2 (C_1 L_1 R_1 g_m + C_2 L_1 R_2 g_m + C_2 L_1) + s (C_2 R_1 R_2 g_m + C_2 R_1 + L_1 g_m)}{C_1 C_2 L_1 s^3 + C_2 s}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 L_2 R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_2 L_1 R_1 + C_2 L_1 L_2 g_m) + s^2 (C_1 L_1 R_1 g_m + C_2 L_1 + C_2 L_2 R_1 g_m) + s (C_2 R_1 + L_1 g_m)}{C_1 C_2 L_1 s^3 + C_2 s}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, L_2 s + R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 L_2 R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_2 L_1 R_1 R_2 g_m + C_1 C_2 L_1 R_1 + C_2 L_1 L_2 g_m) + s^2 (C_1 L_1 R_1 g_m + C_2 L_1 R_2 g_m + C_2 L_1 + C_2 L_2 R_1 g_m) + s (C_2 R_1 R_2 g_m + C_2 R_1 + L_1 g_m)}{C_1 C_2 L_1 s^3 + C_2 s}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s^4 (C_1 C_2 L_1 L_2 R_1 R_2 g_m + C_1 C_2 L_1 L_2 R_1) + s^3 (C_1 L_1 L_2 R_1 g_m + C_2 L_1 L_2 R_2 g_m + C_2 L_1 L_2) + s^2 (C_1 L_1 R_1 R_2 g_m + C_1 L_1 R_1 + C_2 L_2 R_1 R_2 g_m + C_2 L_2 R_1 + L_1 L_2 g_m) + s (L_1 R_2 g_m + L_1 + L_2 R_1 g_m)}{C_1 C_2 L_1 L_2 s^4 + s^2 (C_1 L_1 + C_2 L_2) + 1}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s^4 (C_1 C_2 L_1 L_2 R_1 R_2 g_m + C_1 C_2 L_1 L_2 R_1) + s^3 (C_1 C_2 L_1 R_1 R_2 + C_2 L_1 L_2 R_2 g_m + C_2 L_1 L_2) + s^2 (C_1 L_1 R_1 R_2 g_m + C_1 L_1 R_1 + C_2 L_1 R_2 + C_2 L_2 R_1 R_2 g_m + C_2 L_2 R_1) + s (C_2 R_1 R_2 + L_1 R_2 g_m + L_1)}{C_1 C_2 L_1 L_2 s^4 + C_1 C_2 L_1 R_2 s^3 + C_2 R_2 s + s^2 (C_1 L_1 + C_2 L_2) + 1}$$

$$\mathbf{9.74 \quad X-INVALID-ORDER-74} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_1 C_2 L_1 R_1 s^3 + C_1 L_1 R_1 g_m s^2 + C_2 R_1 s + R_1 g_m}{C_1 C_2 L_1 s^3 + C_1 C_2 R_1 s^2 + C_2 s}$$

9.75 X-INVALID-ORDER-75 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_2 L_1 R_1 R_2 s^3 + C_2 R_1 R_2 s + R_1 R_2 g_m + R_1 + s^2 (C_1 L_1 R_1 R_2 g_m + C_1 L_1 R_1)}{C_1 C_2 L_1 R_2 s^3 + s^2 (C_1 C_2 R_1 R_2 + C_1 L_1) + s (C_1 R_1 + C_2 R_2) + 1}$$

9.76 X-INVALID-ORDER-76 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 R_1 g_m s^2 + R_1 g_m + s^3 (C_1 C_2 L_1 R_1 R_2 g_m + C_1 C_2 L_1 R_1) + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_1 C_2 L_1 s^3 + C_1 C_2 R_1 s^2 + C_2 s}$$

9.77 X-INVALID-ORDER-77 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_2 L_1 L_2 R_1 g_m s^4 + C_1 C_2 L_1 R_1 s^3 + C_2 R_1 s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_2 L_2 R_1 g_m)}{C_1 C_2 L_1 s^3 + C_1 C_2 R_1 s^2 + C_2 s}$$

9.78 X-INVALID-ORDER-78 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, L_2 s + R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_2 L_1 L_2 R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_2 L_1 R_1 R_2 g_m + C_1 C_2 L_1 R_1) + s^2 (C_1 L_1 R_1 g_m + C_2 L_2 R_1 g_m) + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_1 C_2 L_1 s^3 + C_1 C_2 R_1 s^2 + C_2 s}$$

9.79 X-INVALID-ORDER-79 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_2 R_1 g_m s^3 + L_2 R_1 g_m s + R_1 R_2 g_m + R_1 + s^4 (C_1 C_2 L_1 L_2 R_1 R_2 g_m + C_1 C_2 L_1 L_2 R_1) + s^2 (C_1 L_1 R_1 R_2 g_m + C_1 L_1 R_1 + C_2 L_2 R_1 R_2 g_m + C_2 L_2 R_1)}{C_1 C_2 L_1 L_2 s^4 + C_1 C_2 L_2 R_1 s^3 + C_1 R_1 s + s^2 (C_1 L_1 + C_2 L_2) + 1}$$

9.80 X-INVALID-ORDER-80 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \frac{R_2(C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_2 L_1 R_1 R_2 s^3 + C_2 R_1 R_2 s + R_1 R_2 g_m + R_1 + s^4 (C_1 C_2 L_1 L_2 R_1 R_2 g_m + C_1 C_2 L_1 L_2 R_1) + s^2 (C_1 L_1 R_1 R_2 g_m + C_1 L_1 R_1 + C_2 L_2 R_1 R_2 g_m + C_2 L_2 R_1)}{C_1 C_2 L_1 L_2 s^4 + s^3 (C_1 C_2 L_1 R_2 + C_1 C_2 L_2 R_1) + s^2 (C_1 C_2 R_1 R_2 + C_1 L_1 + C_2 L_2) + s (C_1 R_1 + C_2 R_2) + 1}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, L_2 s + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_2 L_1 L_2 R_1 g_m s^2 + C_2 L_1 R_1 s + L_1 R_1 g_m}{C_1 C_2 L_1 R_1 s^2 + C_2 L_1 s + C_2 R_1}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{1}{C_1 R_1}$
 K-LP: $\frac{L_1 g_m}{C_2}$
 K-HP: $\frac{L_2 g_m}{C_1}$
 K-BP: R_1
 Qz: None
 Wz: $\frac{1}{\sqrt{C_2} \sqrt{L_2}}$

10.2 X-INVALID-WZ-2

$$Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 L_1 L_2 R_1 g_m s^2 + L_1 R_1 g_m + s \left(C_2 L_1 R_1 R_2 g_m + C_2 L_1 R_1 \right)}{C_1 C_2 L_1 R_1 s^2 + C_2 L_1 s + C_2 R_1}$$

Parameters:

Q:

$\frac{\sqrt{C_1} R_1}{\sqrt{L_1}}$

wo:

$\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

bandwidth:

$\frac{1}{C_1 R_1}$

K-LP:

$\frac{L_1 g_m}{C_2}$

K-HP:

$\frac{L_2 g_m}{C_1}$

K-BP:

$R_1 R_2 g_m + R_1$

Qz:

None

Wz:

$\frac{1}{\sqrt{C_2} \sqrt{L_2}}$

11 X-PolynomialError